



DUSTSPEC

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DUST-DRIVEN VERTICAL SHEAR INSTABILITY

Matthijsse J.; Aly H. & Paardekooper S.-J. (In prep.)

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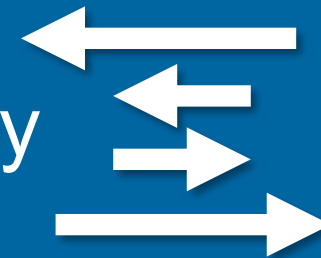
Using Simulations, we investigate how the Dust-Driven Vertical Shear Instability (DVSI) drives Turbulence and assess its Efficiency in Vertically Mixing Dust.

Vertical Shear Instability

Urpin & Brandenburg (1998); Nelson et al. (2013)

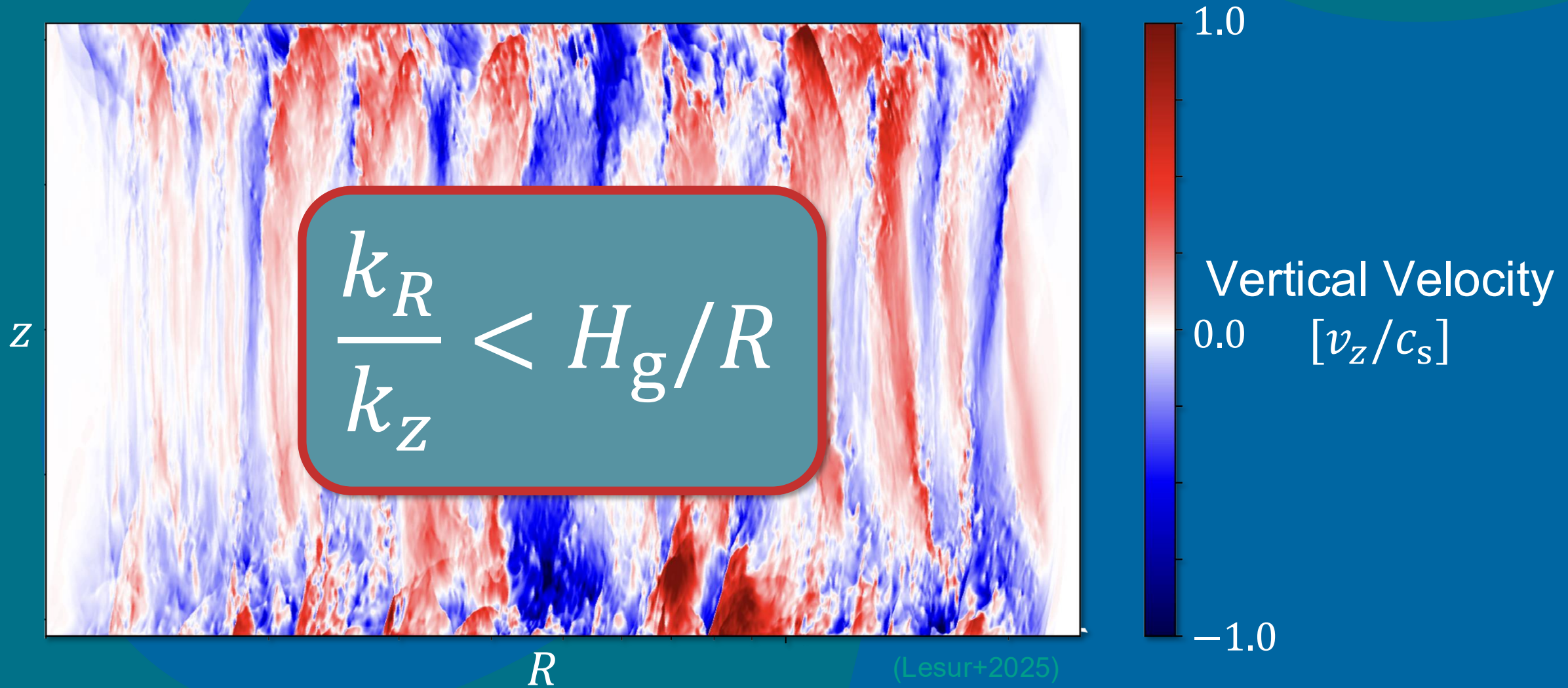


Free Energy from Shear Velocity



$$\frac{\partial(R\Omega)}{\partial z} \neq 0$$

Vertical Shear Instability



Global (Baroclinic)

Temperature structure

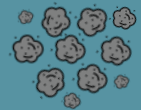
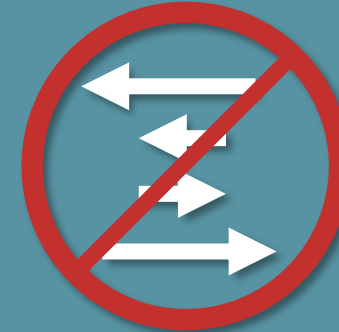
$$\frac{\partial(R\Omega)}{\partial z} \neq 0$$



Local (Barotropic)

Globally isothermal

$$\frac{\partial(R\Omega)}{\partial z} = 0$$



Dust Traps



Temperature



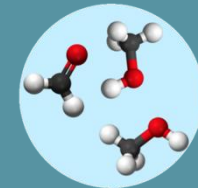
Stabilised by Buoyancy



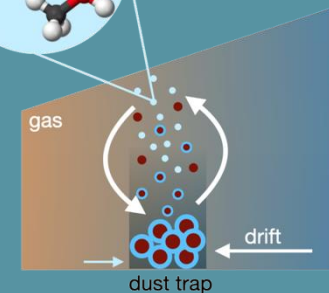
Chemistry

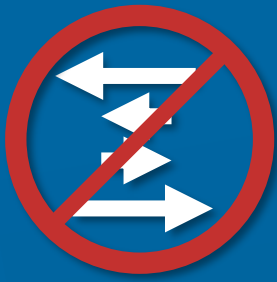


Vertical mixing



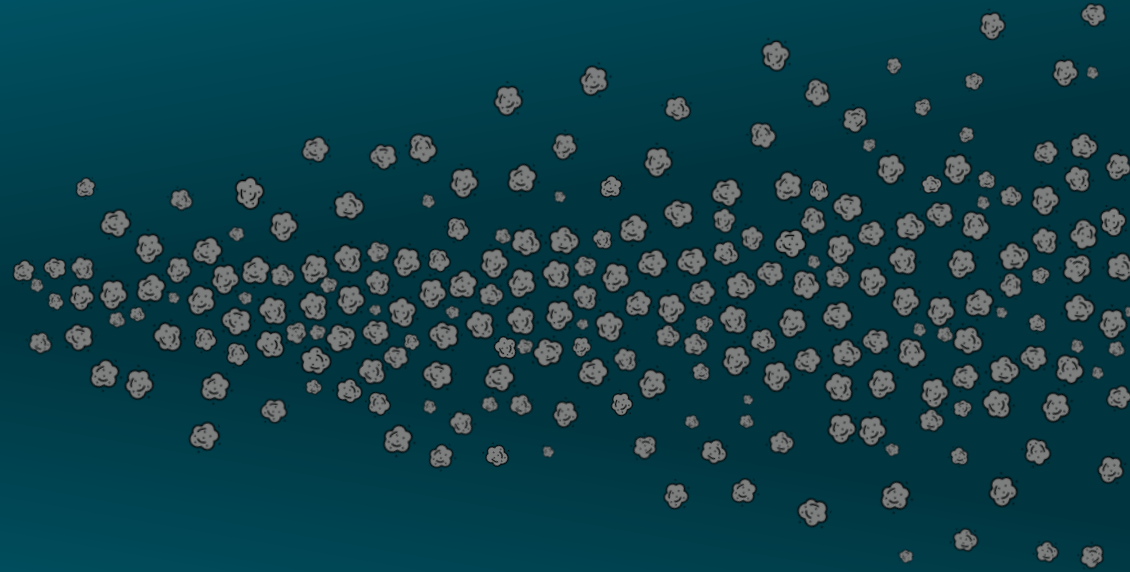
IRS-48



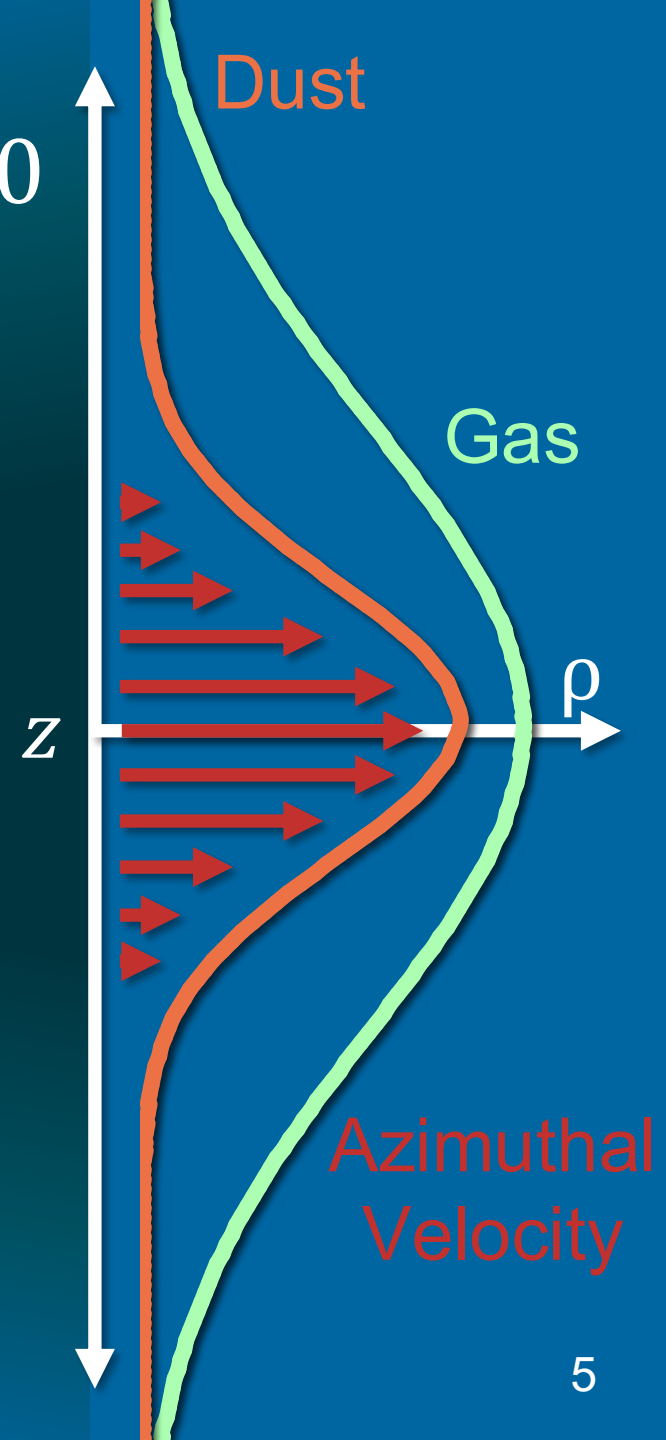


Barotropic Disc

$$\frac{\partial(R\Omega)}{\partial z} = 0$$



$$\Omega_{\text{Dust}} > \Omega_{\text{Gas}}$$



Equilibrium Profile

Vertical shear drives the Instability

Vertical Shearing
Streaming Instability
(VSSI; [Lin 2021](#))

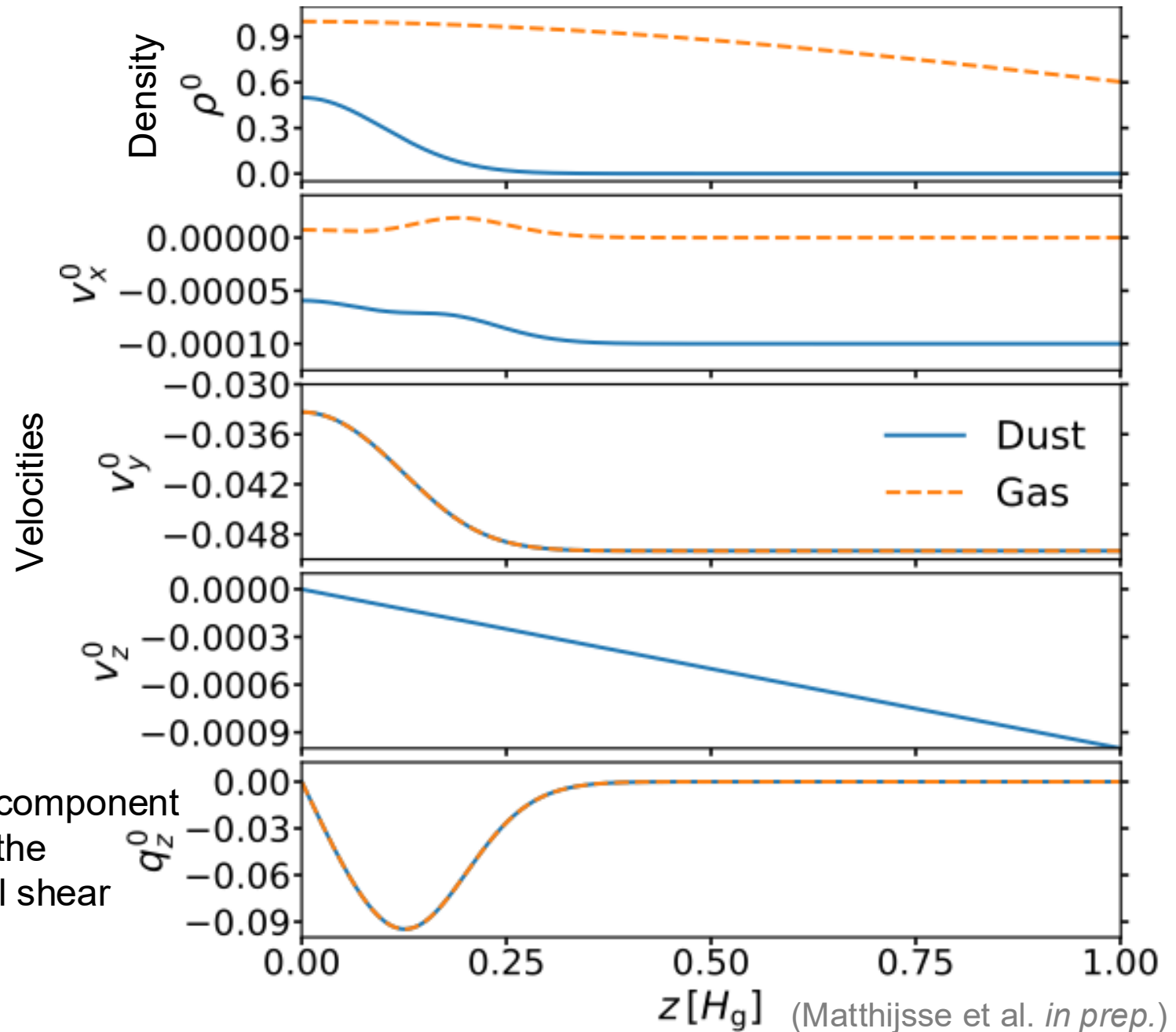
Symmetric Instability
(SymI; [Sengupta & Umurhan 2023](#))

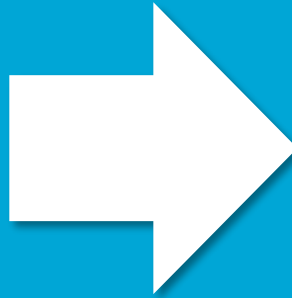
Drag Force

$$\mathbf{a}_{g, \text{drag}} = \frac{\rho_d}{\rho_g} \frac{\mathbf{v}_d - \mathbf{v}_g}{\tau_s}$$

Azimuthal component
of the
vertical shear

$$Z = 0.05; \tau_s = 10^{-3}; \alpha_D = 10^{-5}; \alpha_g = 0; \eta = 0.05$$





Unstratified

Stratified

2D*

3D

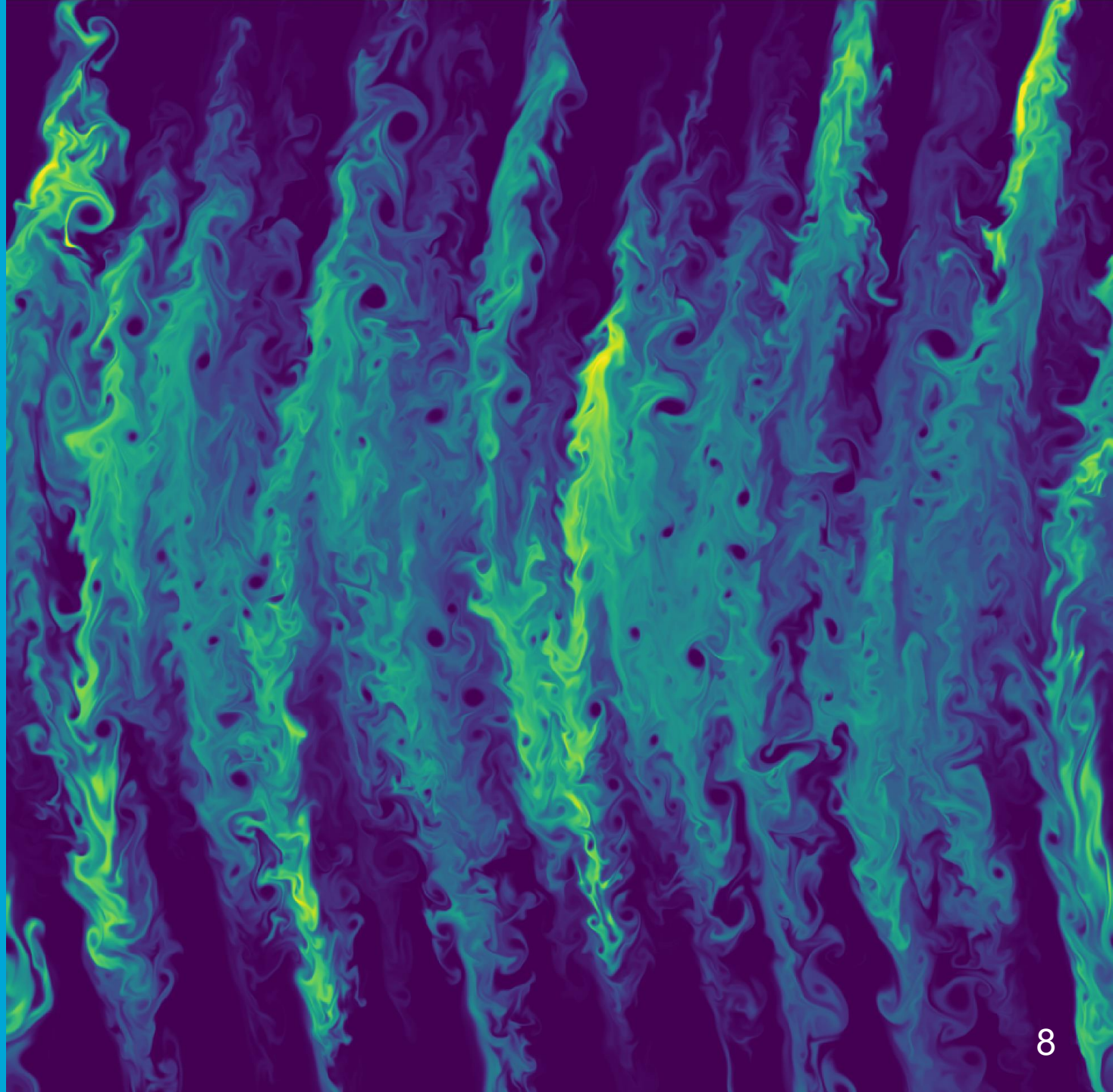
No Feedback**

Dust Feedback

*Axisymmetric (R-z) the “classical” Kelvin-Helmholtz Instability cannot grow

**We use a “Dust-analogue”, the drag on the gas $a_{g, \text{drag}}^0$ is constant.

Results

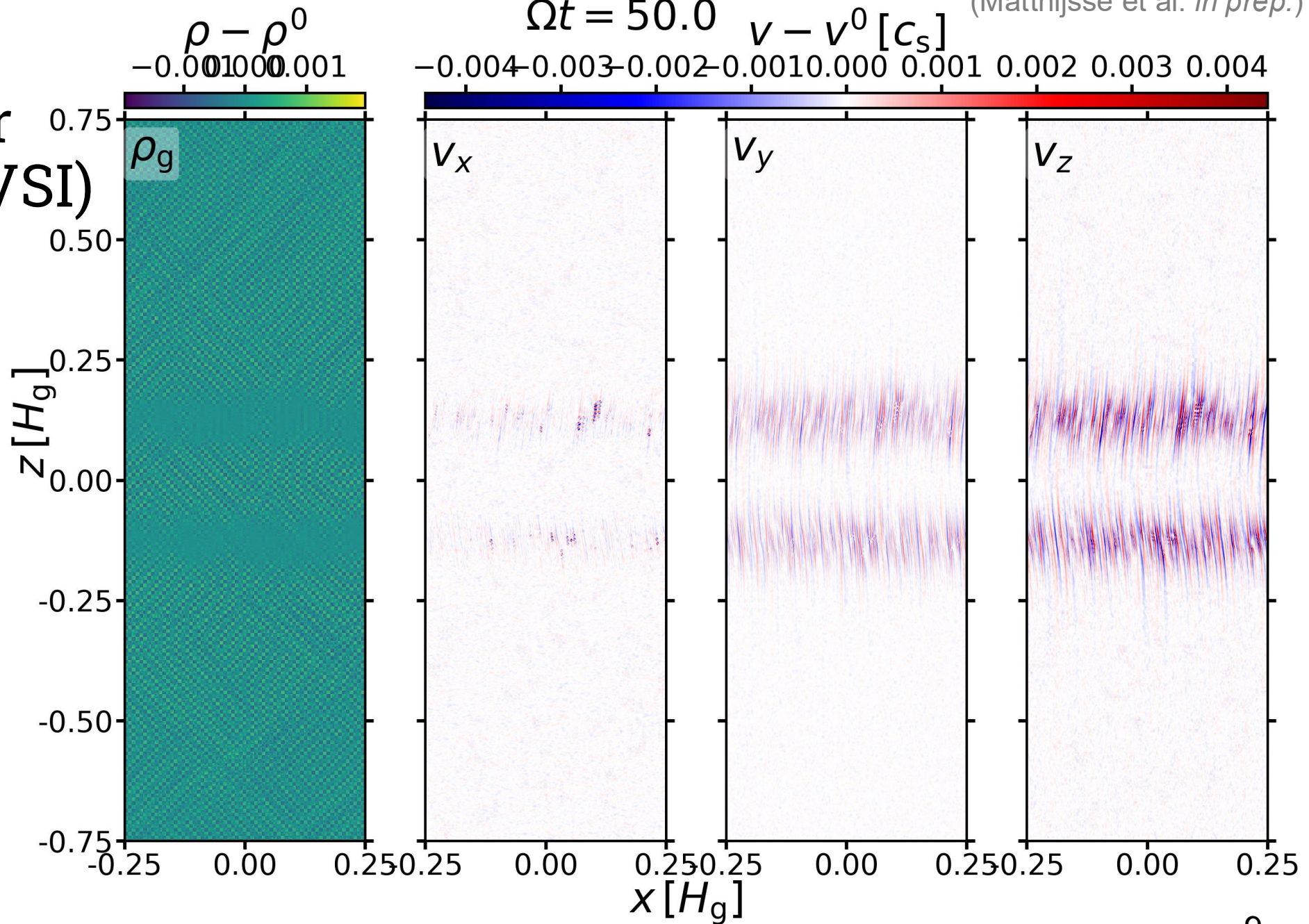


Dust-Driven

Vertical Shear Instability (DVSI)

Heatmap: perturbed gas
density and velocity

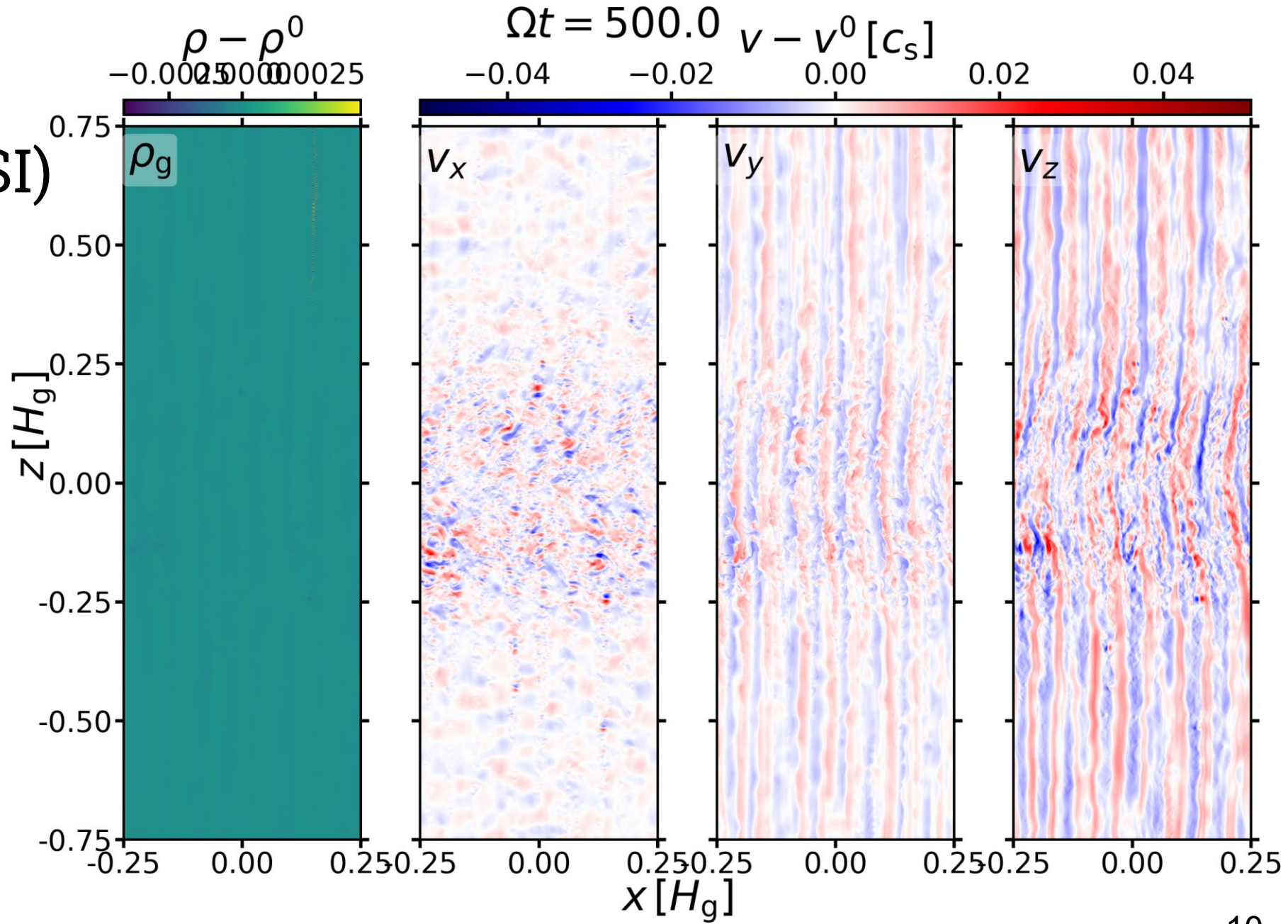
$Z = 0.05$
 $\tau_s = 10^{-3}$
 $\alpha_D = 10^{-5}$
 $\alpha_g = 0$
No Feedback



Dust-Driven Vertical Shear Instability (DVSI)

Heatmap: perturbed gas
density and velocity

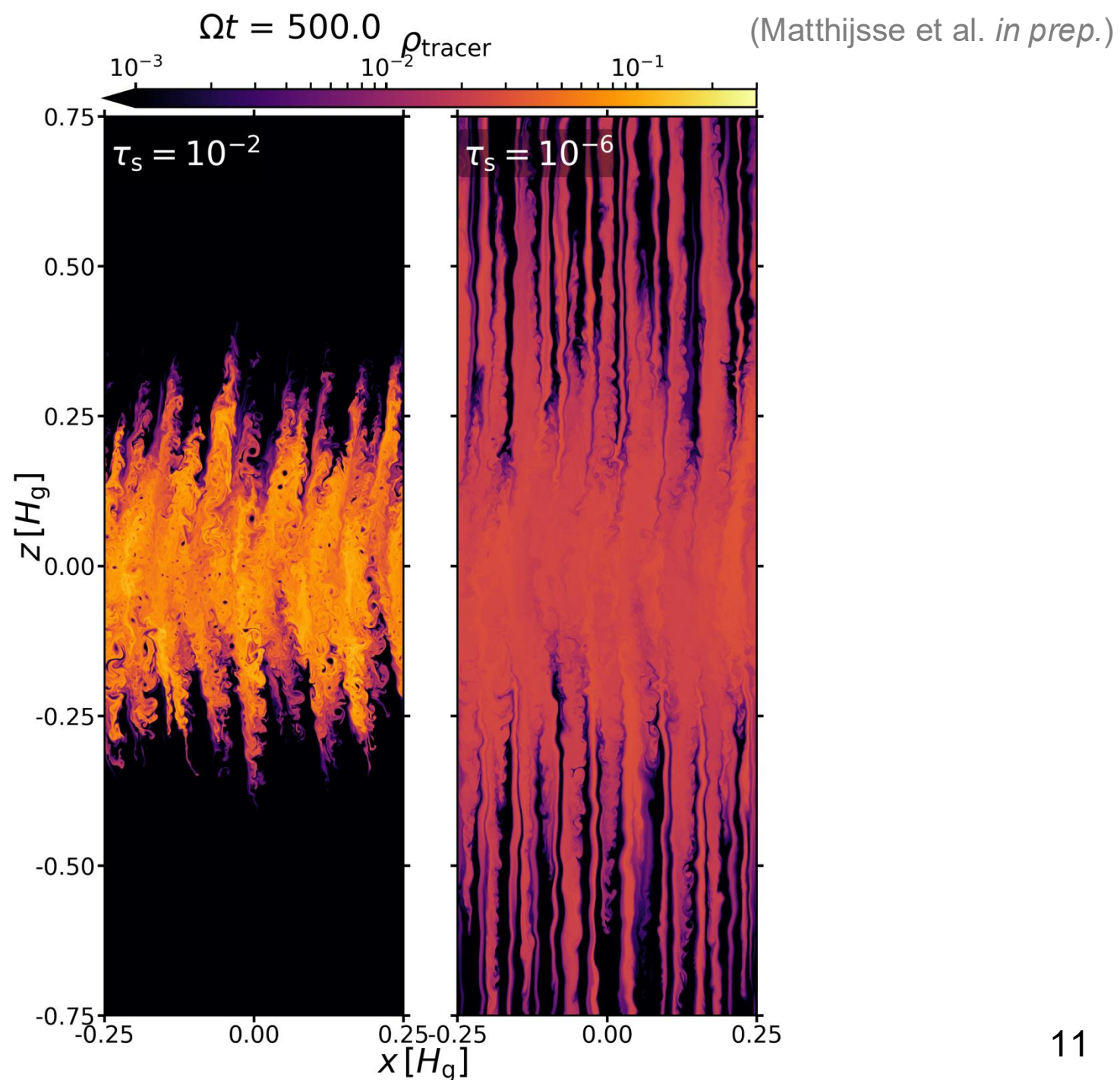
$Z = 0.05$
 $\tau_s = 10^{-3}$
 $\alpha_D = 10^{-5}$
 $\alpha_g = 0$
No Feedback



Dust-Driven Vertical Shear Instability (DVSI)

Heatmap: density of tracer fluids

$Z = 0.05$
 $\tau_s = 10^{-3}$
 $\alpha_D = 10^{-5}$
 $\alpha_g = 0$
No Feedback



Reynolds stress tensor

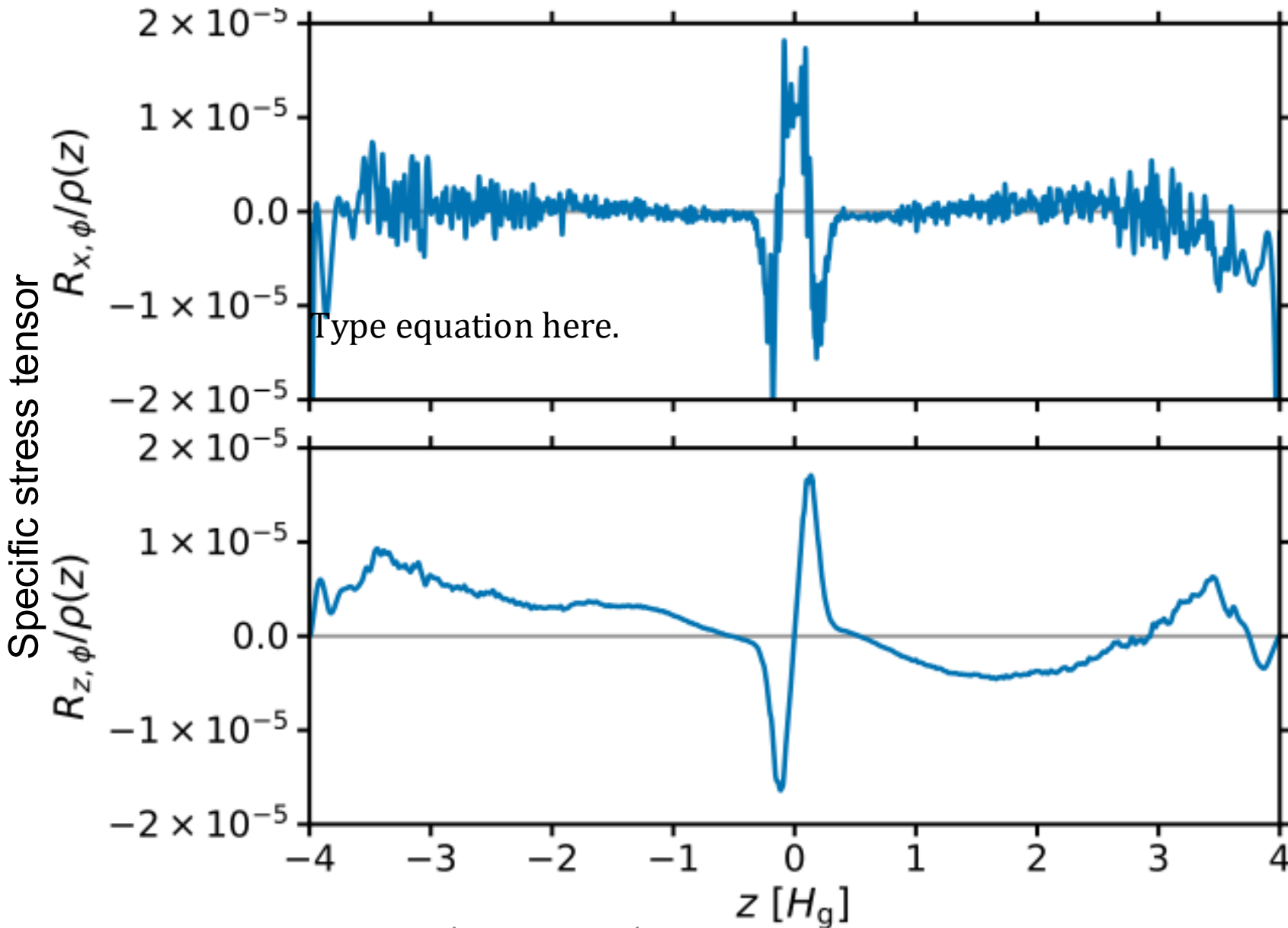
Isotropic turbulence near the midplane

Anisotropic turbulence at higher scale heights

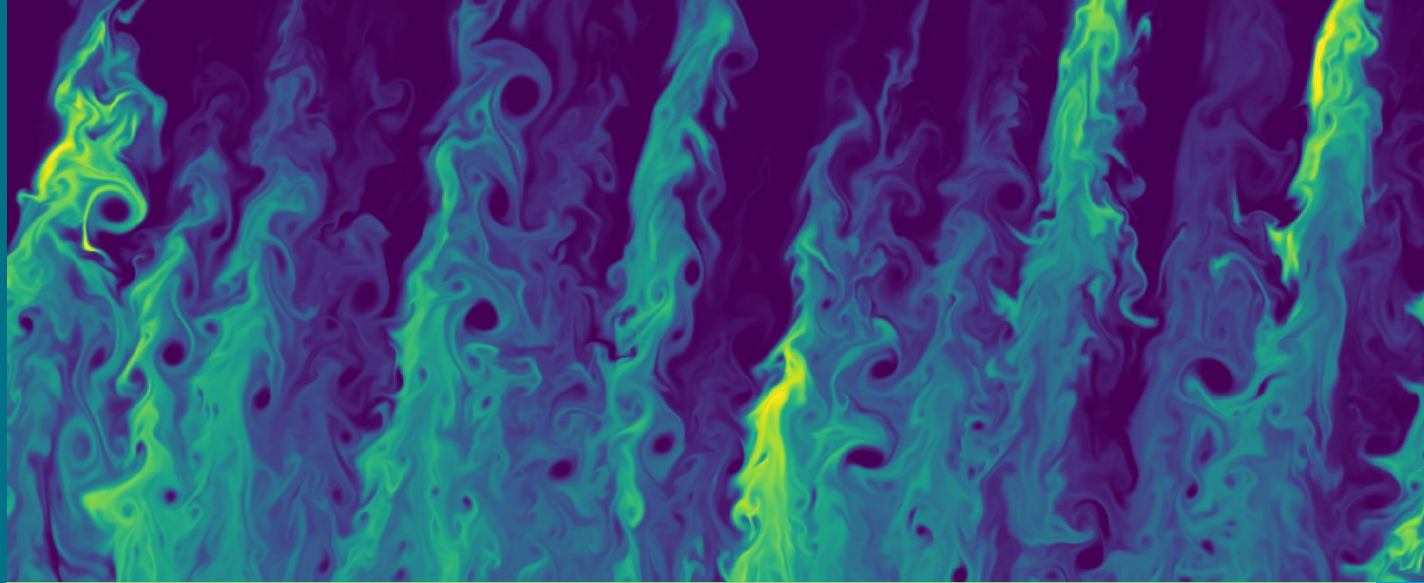
$Z = 0.05$
 $\tau_s = 10^{-3}$
 $\alpha_D = 10^{-5}$
 $\alpha_g = 0$
No Feedback



$\eta = 0.05; N_{\text{grid}} = 1024 \text{ px}/H_g; L_{R,z} = [1 \times 8] H_g$



$$R_{i,\phi}(z) = \langle \rho_g u_i \delta u_\phi \rangle_{t,\phi,R}$$



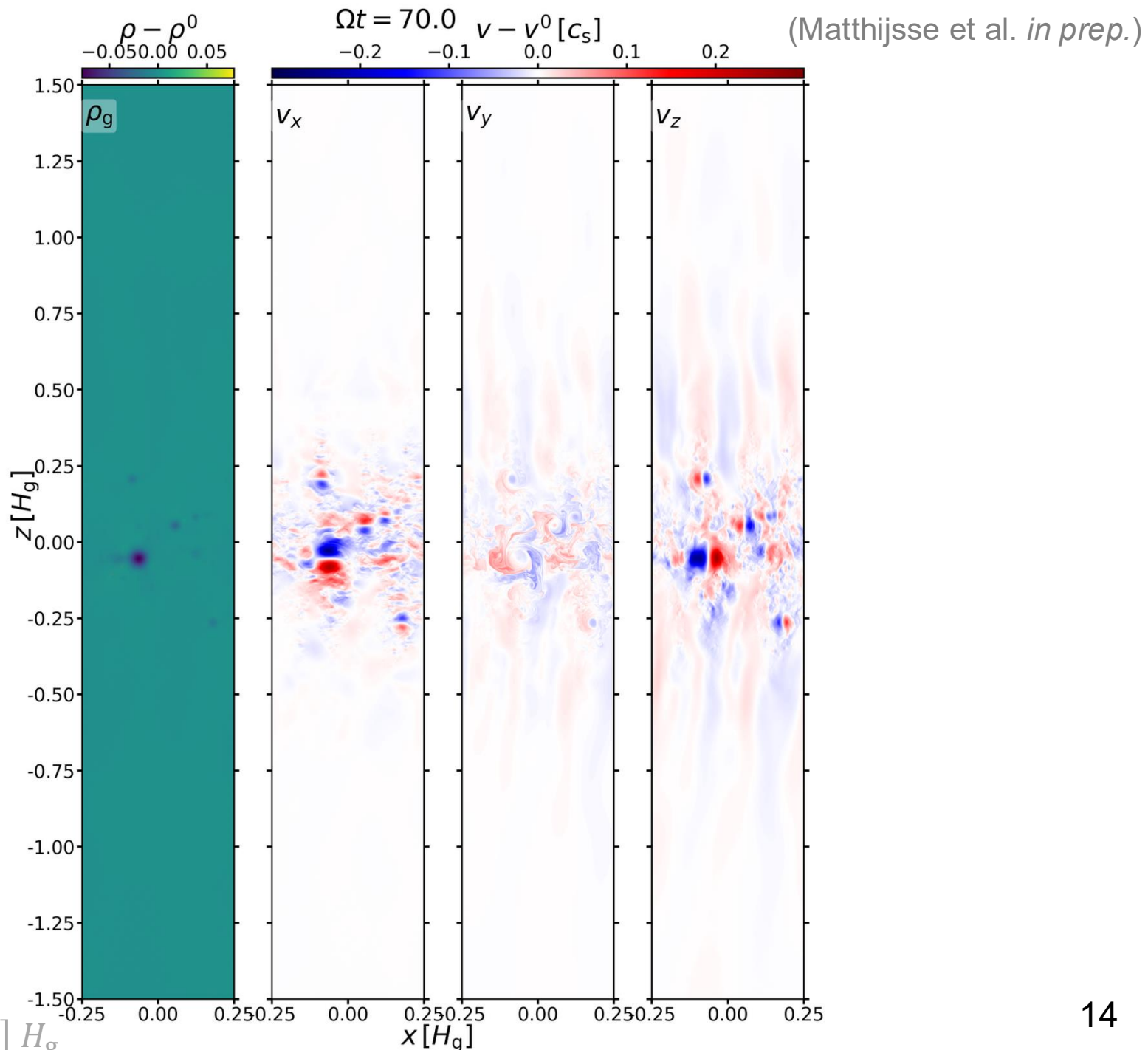
Kelvin Helmholtz

Thin shear profile

Kelvin-Helmholtz Like Instability

Heatmap: perturbed gas
density and velocity

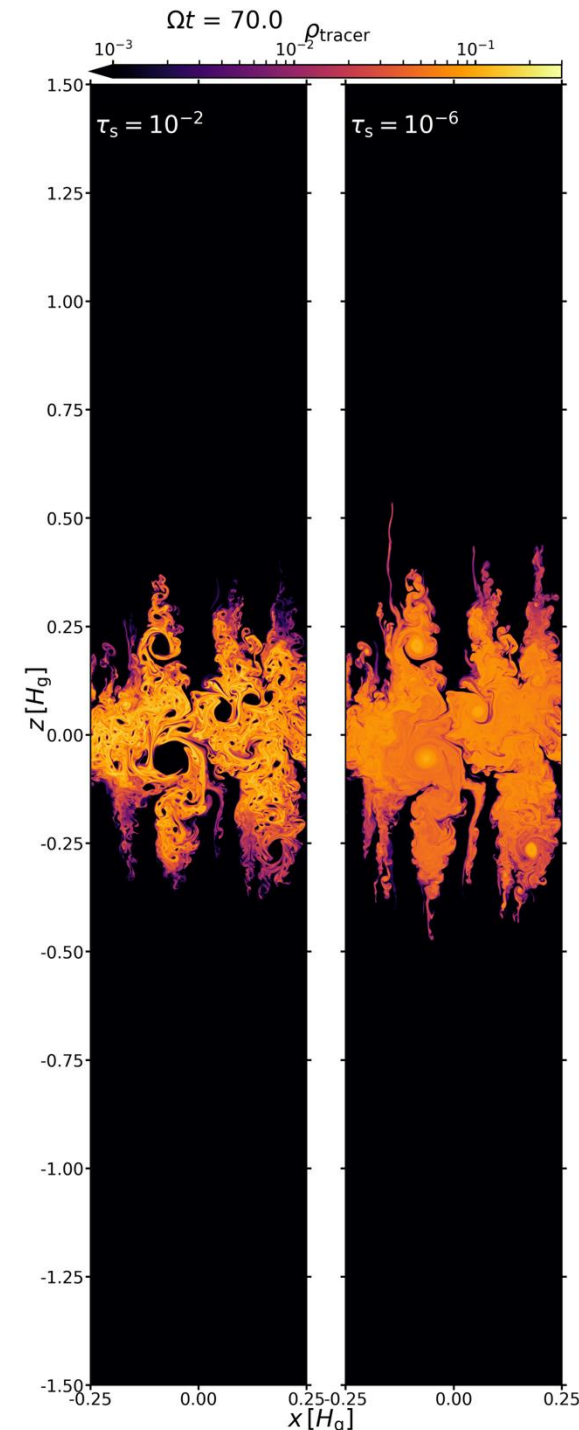
$Z = 0.05$
 $\tau_s = 10^{-3}$
 $\alpha_D = 10^{-6}$
 $\alpha_g = 0$
No Feedback



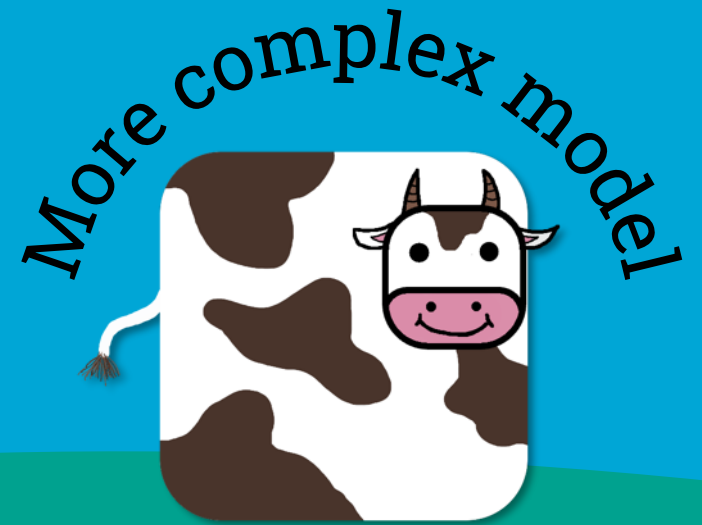
Kelvin-Helmholtz Like Instability

Heatmap: density of tracer fluids

$Z = 0.05$
 $\tau_s = 10^{-3}$
 $\alpha_D = 10^{-6}$
 $\alpha_g = 0$
No Feedback



(Matthijsse et al. *in prep.*)



Unstratified

Stratified

2D

3D

No Feedback*



Dust Feedback



*We use a "Dust-analogue", the drag on the gas $a_{g,drag}^0$ is constant.

Vertical Shearing Streaming Instability (VSSI)

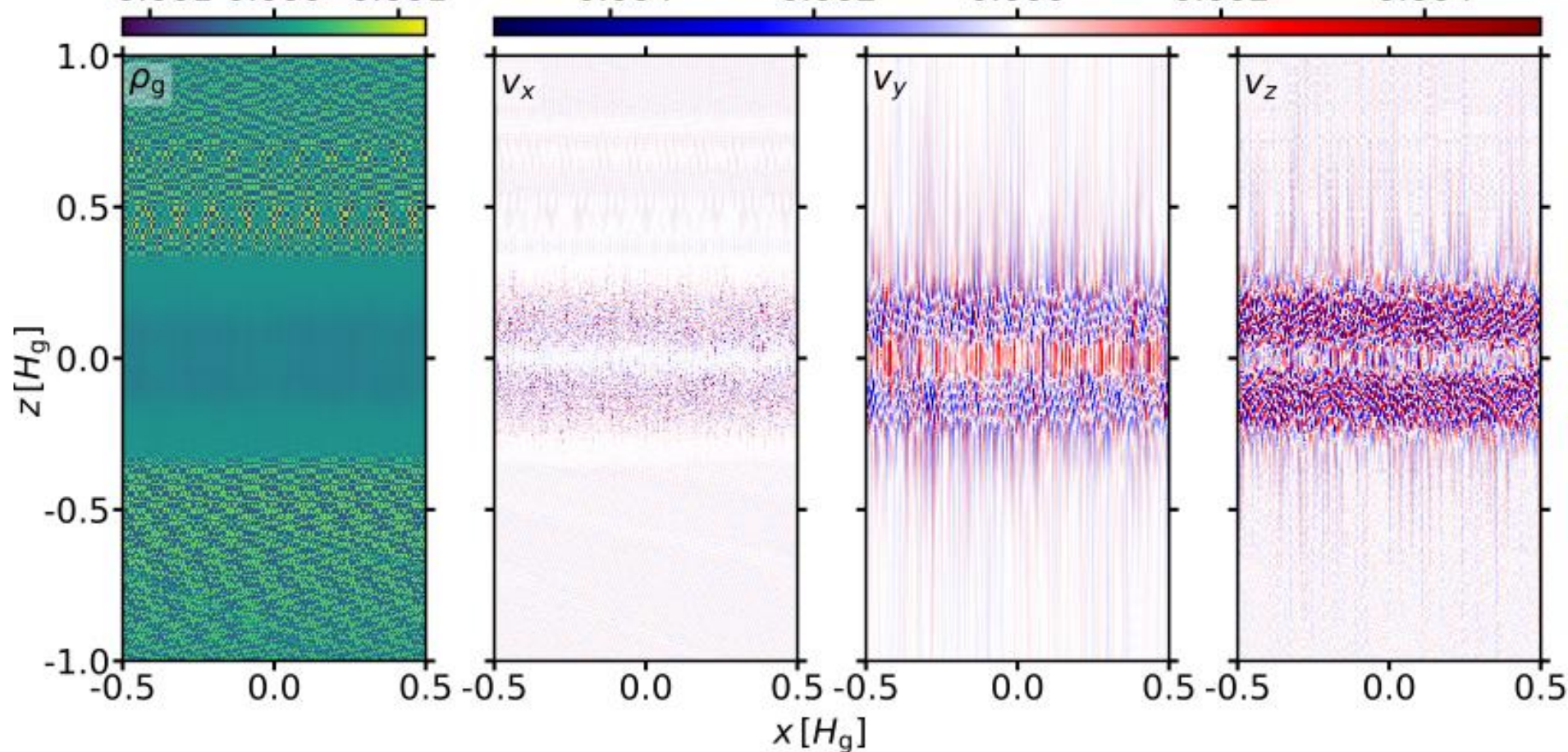
The role of feedback

Heatmap: perturbed gas density and velocity

Perturbed density $[\rho_g - \rho_g^0]$
-0.001 0.000 0.001

Perturbed velocity $[v - v^0]$
-0.004 -0.002 0.000 0.002 0.004

Time $\Omega t = 500$



$$Z = 0.05$$

$$\tau_s = 10^{-3}$$

$$\alpha_D = 10^{-5}$$

$$\alpha_g = 0$$

Feedback

DVSI vs VSSI: The role of feedback

Time evolution: scale height tracer fluids
& direction component kinetic energy gas

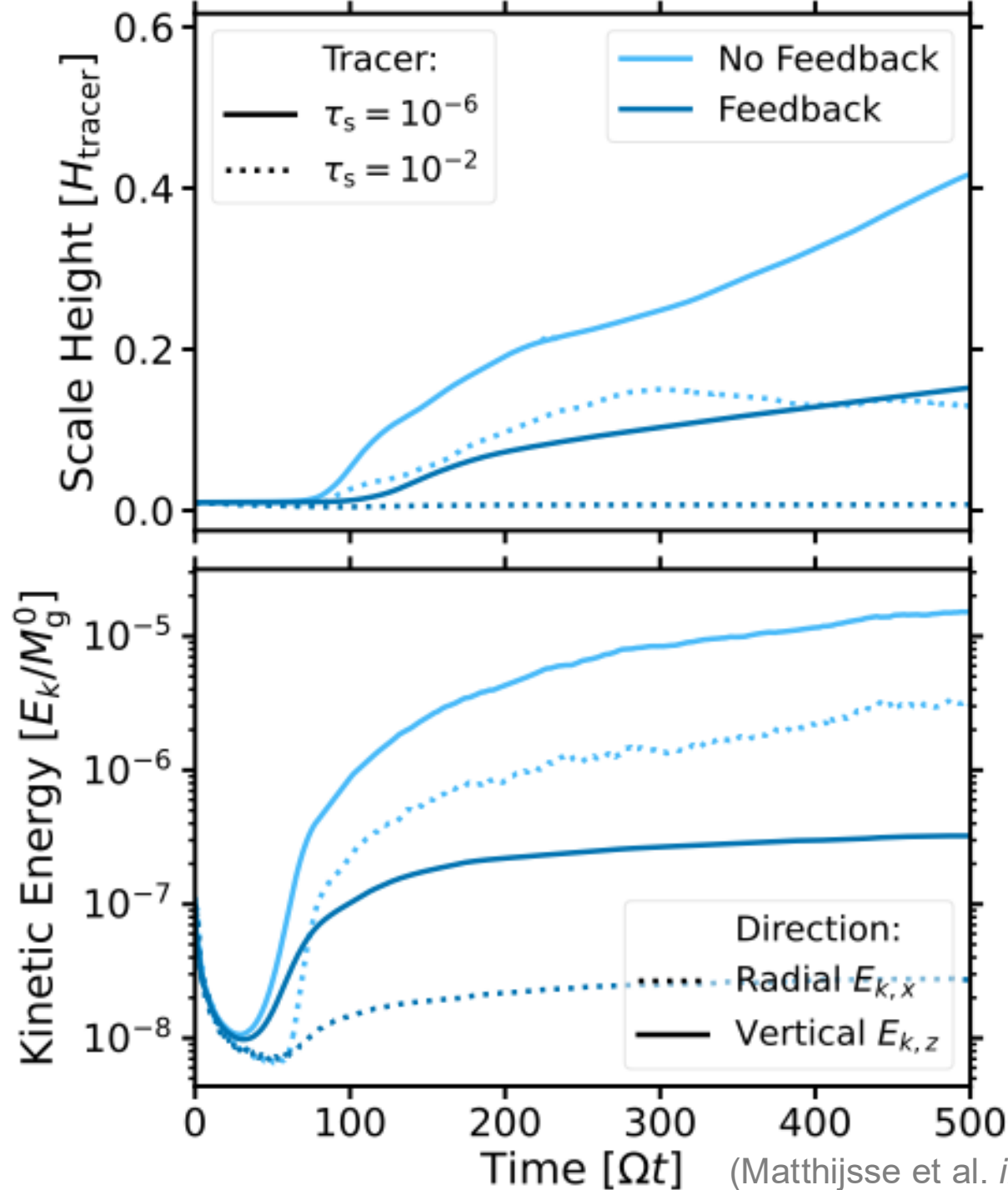
$$Z = 0.05$$

$$\tau_s = 10^{-3}$$

$$\alpha_D = 10^{-5}$$

$$\alpha_g = 0$$

No Feedback



Changing the dust layer

Equilibrium profile: directional component vertical shear gas

$$\begin{aligned} Z &= 0.05 \\ \tau_s &= 10^{-3} \\ \alpha_D &= 10^{-5} \\ \alpha_g &= 0 \end{aligned}$$

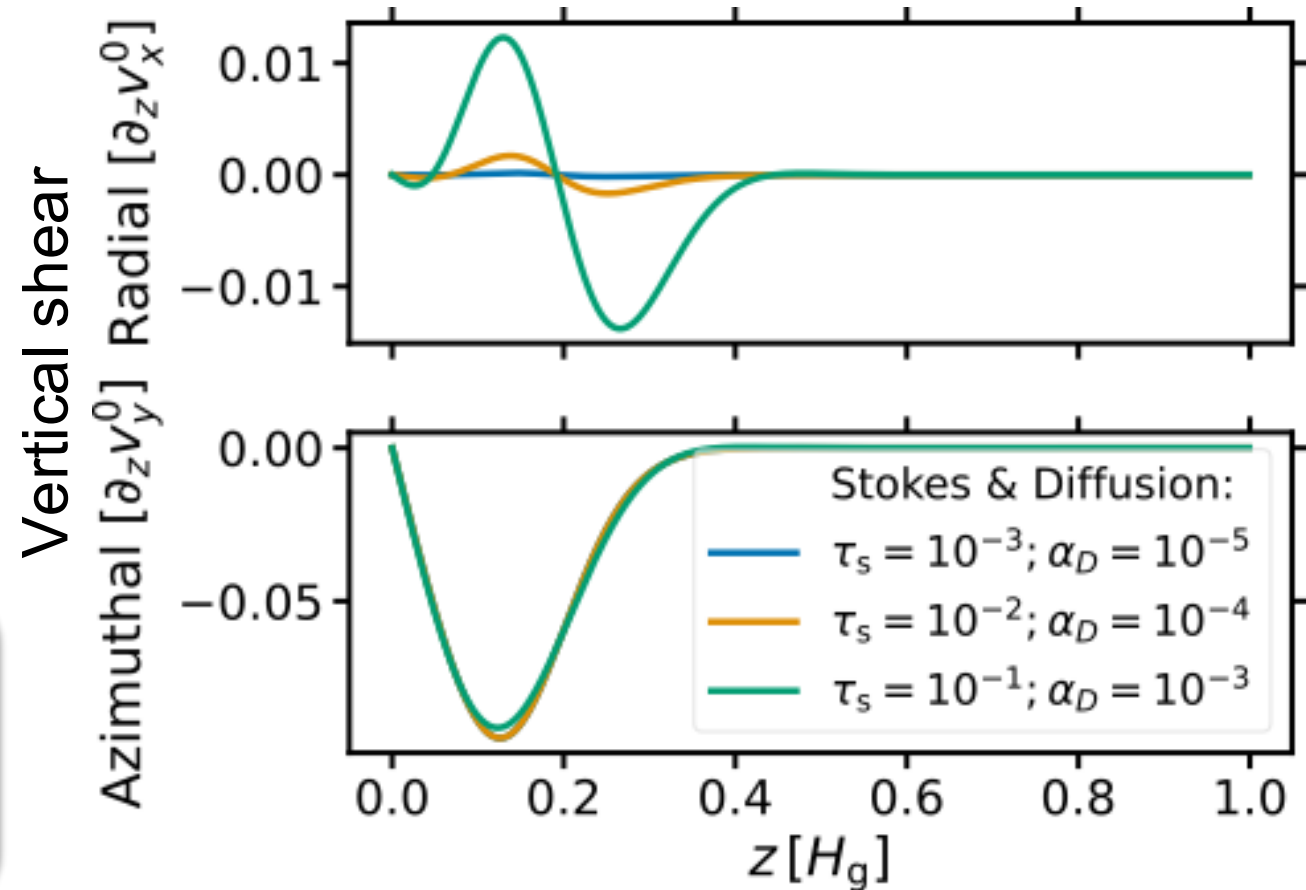
$$\begin{aligned} Z &= 0.05 \\ \tau_s &= 10^{-2} \\ \alpha_D &= 10^{-4} \\ \alpha_g &= 0 \end{aligned}$$

$$\begin{aligned} Z &= 0.05 \\ \tau_s &= 10^{-1} \\ \alpha_D &= 10^{-3} \\ \alpha_g &= 0 \end{aligned}$$

Less coupled dust layer

Changing the Stokes number & dust diffusion together will not affect the azimuthal vertical shear

The dust layer with larger Stokes number and dust diffusion is less coupled to the gas



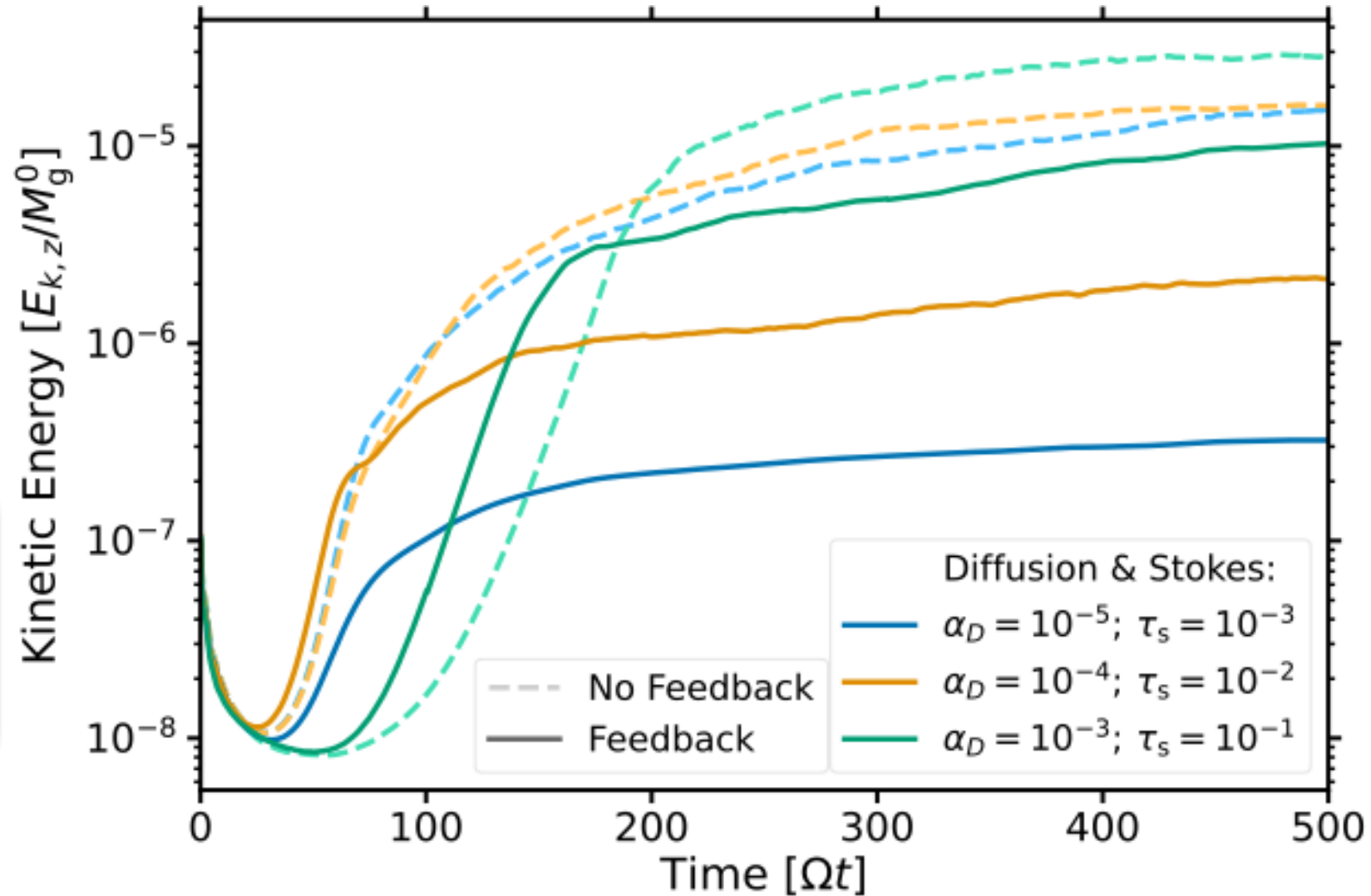
Changing the dust layer

Time evolution: vertical component
kinetic energy gas

$Z = 0.05$
 $\tau_s = 10^{-3}$
 $\alpha_D = 10^{-5}$
 $\alpha_g = 0$
No Feedback

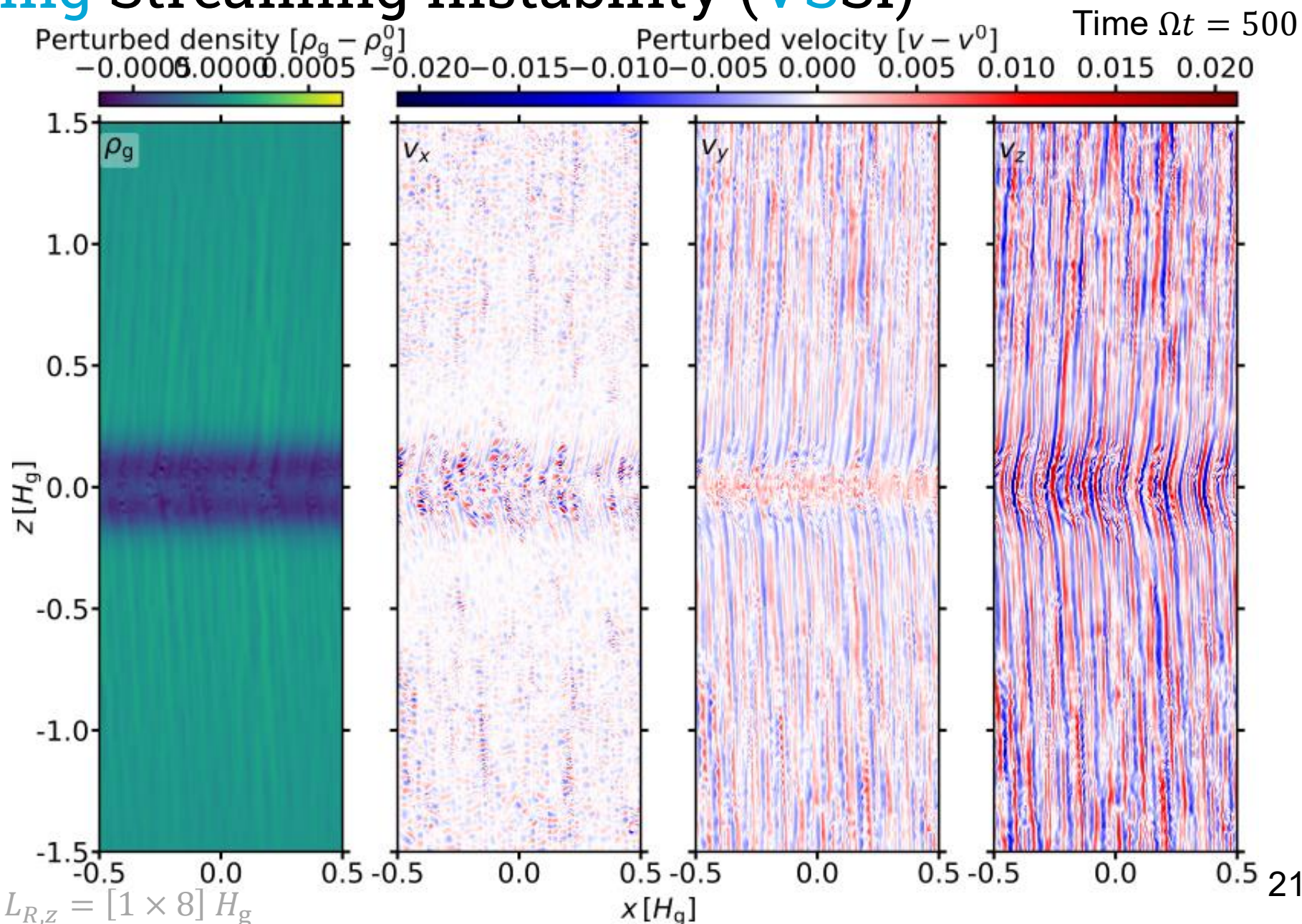
$Z = 0.05$
 $\tau_s = 10^{-2}$
 $\alpha_D = 10^{-4}$
 $\alpha_g = 0$
No Feedback

$Z = 0.05$
 $\tau_s = 10^{-1}$
 $\alpha_D = 10^{-3}$
 $\alpha_g = 0$
No Feedback



Vertical Shearing Streaming Instability (VSSI)

Heatmap: perturbed gas density and velocity



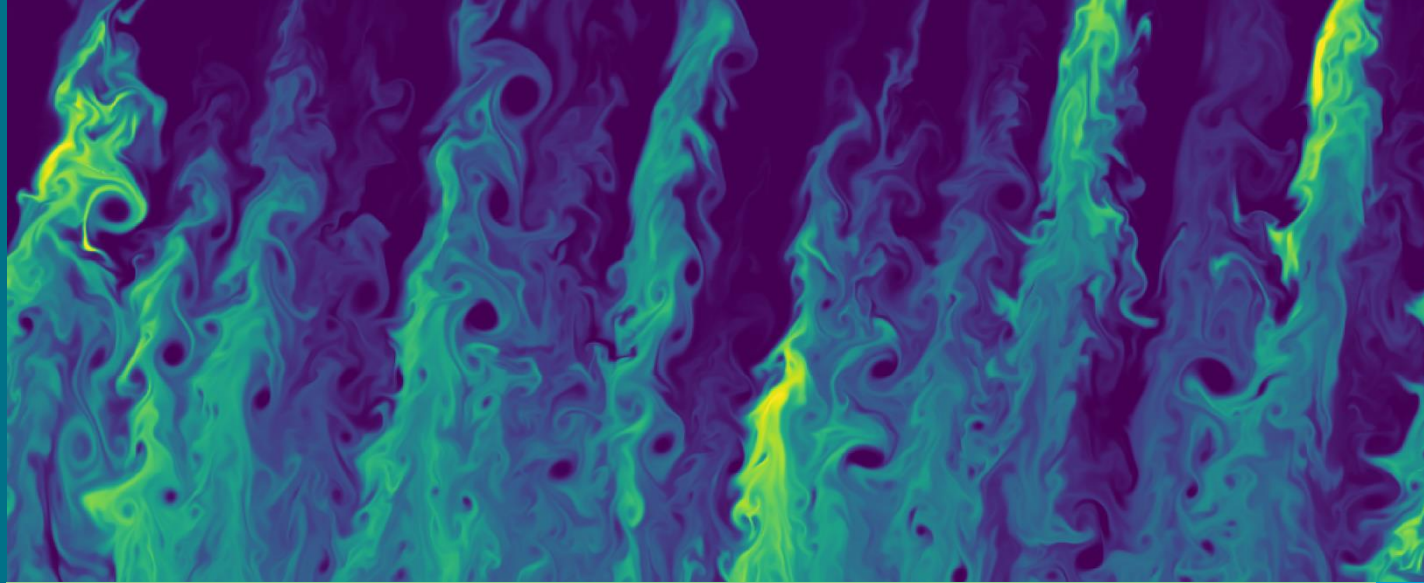
$$Z = 0.05$$

$$\tau_s = 10^{-1}$$

$$\alpha_D = 10^{-3}$$

$$\alpha_g = 0$$

Feedback

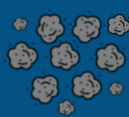


Observations

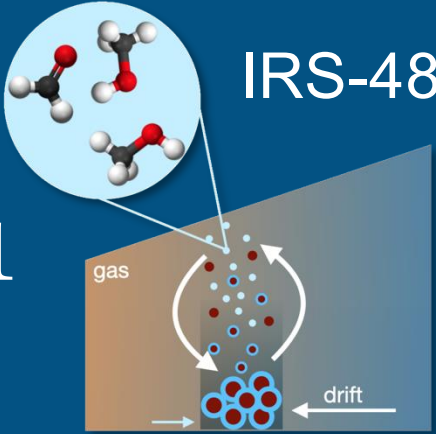
IRS-48

Vertical Mixing in IRS-48

Temmink et al. (2025); Leemker et al. (2023); van der Marel et al. (2021)



Dust Traps



IRS-48

Turbulence $\alpha_D > 0.01$

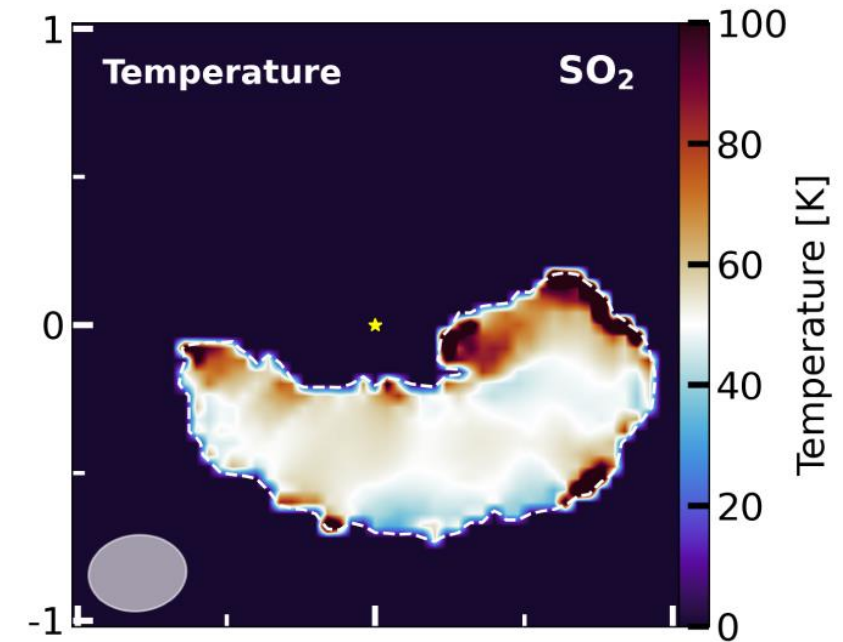


Chemistry





Vertical Mixing



- ★ The DVSI drives extended **vertical modes** (large k_x/k_z)
- ★ Near the midplane, the **secondary** instability stabilizes the DVSI
- ★ There is a regime **change** between the DVSI and Kelvin-Helmholtz roll-up

This depends on the strength of the **Radial** Vertical Shear

- ★ Dust **feedback** dampens the DVSI
- ★ The DVSI can drive strong vertical **mixing** in the dust trap of IRS-48

